Answer key: Lesson 2, Number Words and the Number Line

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Write the missing numbers on the number line.

Answer: 10 and 12

The line goes 8, 9, 10, 11, 12, 13, 14. The empty boxes are 10 and 12.

Watch for: Writing 9 or 11 again (not counting on).; Leaving a box empty.

S2. Write the missing numbers.

Answer: 29; 39 and 41; 70

(a) 28, then 29, then 30. (b) 38, then 39, then 40, then 41: the line goes on after 40. (c) One before 71 is 70.

Watch for: Writing 310 for the number after 38 (the ones digit goes past 9).; Stopping at 40 and leaving the last box empty.

S3. Write the number one before and the number one after.

Answer: 34 and 36; 59 and 61; 78 and 80

(a) 34 and 36. (b) Back from 60 is 59, on from 60 is 61. (c) Back from 79 is 78. After 79 the ones are used up, so a new ten begins: 80.

Watch for: Writing 50 and 70 (a whole ten instead of one step).; Writing 710 for the number after 79.

S4. Read the number word. Write the number.

Answer: 15, 50, 24

(a) fifteen is ten and 5: 15. (b) fifty is 5 tens: 50. (c) twenty is 20 and 4 more: 24.

Watch for: Writing 50 for fifteen, or 15 for fifty (teen and ty mixed up).; Writing 42 for twenty-four (digits the wrong way round).

S5. Copy the number word. Word bank: twenty, forty, sixty.

Answer: twenty, forty

20 is 2 tens: twenty. 40 is 4 tens: forty (no u).

Watch for: Writing fourty (forty has no u).; Copying the word for a different number, for example sixty for 40.

S6. This number line counts in tens. Write the missing numbers.

Answer: 30, 50 and 80

0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. The missing marks are 30, 50 and 80.

Watch for: Writing 21, 25 or 75 (counting on in ones instead of tens).; Leaving out the mark at 80.

S7. Which two tens is each number between?

Answer: 20 and 30; 60 and 70

(a) 23 is 2 tens and 3 more: between 20 and 30. (b) 68 is between 60 and 70.

Watch for: Writing 2 and 3 (digits) instead of 20 and 30.; Choosing the ten below twice, for example 60 and 60.

S8. Siti wrote 57, 58, 59, 510, 511. What should come after 59? Say what she mixed up.

Answer: 60. After 9 the ones are used up, so a new ten begins. Siti kept counting ones.

After 59 the ones are used up, so we start a new ten: 6 tens and 0 ones, 60. Siti kept adding ones after the 9, so she wrote 510.

Watch for: Writing 510 (the ones digit goes past 9).; Not saying what she mixed up (any clear sentence about the tens is fine).

Think harder (page 3)

T1. Which ten is nearer? Count the hops on the number line to check.

Answer: 50; 90

(a) 47 to 50 is 3 hops, 47 to 40 is 7 hops: 50. (b) 89 to 90 is 1 hop, 89 to 80 is 9 hops: 90.

Watch for: Choosing the ten with the same tens digit (40 for 47) without counting the hops.; Counting the hops from the wrong ten.

T2. How many numbers are between 28 and 35? Do not count 28 or 35.

Answer: 6 (29, 30, 31, 32, 33, 34)

29, 30, 31, 32, 33, 34. That is 6 numbers. The hops from 28 to 35 are 7, but the numbers in the middle are one fewer.

Watch for: Answering 7 (counting the hops, or counting 35).; Answering 8 (counting 28 and 35 as well).

T3. Mystery number. Clue 1: I am between 50 and 60. Clue 2: I am nearer to 60 than to 50. Clue 3: One before me is halfway between 50 and 60. What number am I?

Answer: 56

Between 50 and 60 and nearer to 60: 56, 57, 58 or 59. The halfway number is 55. One after 55 is 56, so clue 3 fits only 56.

Watch for: Answering 55 (that is halfway, not nearer to 60).; Answering 54 (one before 55: clue 3 says one before ME is halfway).

T4. The frog is on 96. It hops on one number at a time to 100. Write each number it lands on, and how many hops.

Answer: 97, 98, 99, 100 (4 hops)

97, 98, 99, 100. The frog makes 4 hops. After 99 the ones and the tens are used up, so we write 100, one hundred.

Watch for: Writing 910 or 9100 after 99.; Writing 96 as a landing (the frog starts there).

Go further (page 4)

G1. The boxes on each line are equal jumps. Fill in the boxes and write how big each jump is.

Answer: (a) 10, 20, jump 10. (b) 15, 20, 25, jump 5. (c) 53, 56, 59, jump 3.

(a) 3 jumps from 0 to 30: 10 each. (b) 4 jumps from 10 to 30, which is 20 altogether: 5 each. (c) 4 jumps from 50 to 62, which is 12 altogether: 3 each.

Watch for: Always using a jump of 10.; Counting the boxes (3) as the jumps (4).

G2. Mei's frog is on 28. Ali's frog is on 52. They hop towards each other, one number at a time, at the same speed. On which number do they meet?

Answer: 40

From 28 to 52 is 24 hops (28 to 30 is 2, 30 to 50 is 20, 50 to 52 is 2). Each frog makes half: 12 hops. 28 and 12 more is 40. 52 and 12 back is 40. They meet on 40.

Watch for: Choosing 38 or 50, one frog's own number, or adding 28 and 52.; Counting the hops wrongly across the tens.

G3. Make your own. Draw a number line with 5 boxes in a row. Fill in two boxes that are not next to each other, so that only one number fits each empty box. Give it to a grown-up to solve.

Answer: Any correct line, for example 15, [], [], [], 35 (jump 5: 20, 25, 30).

Example: jump 5: 15, 20, 25, 30, 35. Hide 20, 25 and 30. The grown-up counts the 4 jumps from 15 to 35, which is 20 altogether, so the jump is 5, and the boxes are 20, 25 and 30. Check: the same jump between every two boxes.

Watch for: Making jumps that are not equal.; Giving two numbers that do not fit one equal jump.

Investigation (page 5): Halfway

41, 42, 43 and 44 are nearer to 40. 46, 47, 48 and 49 are nearer to 50. 45 is exactly halfway: 5 hops each way. Rule: ones digit 1 to 4, the lower ten is nearer; ones digit 6 to 9, the higher ten is nearer; ones digit 5, halfway. It works between every two tens, for example 61 to 69.

Look for: A child who tests several numbers by counting hops, notices 45 is special, and says why the ones digit decides it. A child who finds the same pattern between 60 and 70 has generalised.

Exit ticket (page 5)

E1. Read the number word. Write the number.

Answer: 73

Seventy is 70, and 3 more makes 73.

Watch for: Writing 37.; Writing 703.

E2. Which two tens is 76 between?

Answer: 70 and 80

76 is 7 tens and 6 more, so it is past 70 and not yet 80.

Watch for: Writing 7 and 8.; Writing 60 and 70.

E3. The frog is on 99. It hops one number forward. Write where it lands.

Answer: 100

After 99 comes 100, one hundred: ten tens.

Watch for: Writing 910 or 9100.; Writing 90.

Sneak-peek puzzles (not in the workbook)

1. 20 and 23 stars.
2. 40. From 38 to 40 is 2 hops. From 38 to 30 is 8 hops.